MILITARY SPECIFICATION

BEARING, BALL, ANNULAR, ANTIFRICTION, ACCESSORY, HIGH SPEED TURBINE, AIRCRAFT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers high speed ball bearings intended primarily for use in aircraft accessory turbines.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-A-624	Steel Bar, Alloy, Hot Rolled and Cold Finished, General Purpose

Military

MIL-B-197	Bearing, Anti-friction, Associated Parts and Subassemblies, Packaging of
MIL-B-1083	Ball, Bearing, Ferrous and Non-Ferrous
MIL-I-6866	Inspection, Penetrant Method of
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-S-7420	Steel Bars, Alloy, Chromium, High Carbon E52100, and E51100 (Aircraft Quality)

STANDARDS

Federal

FED. STD.NO. 66	Steel: Chemical Composition and Hardenability
FED. TEST METHOD STD. NO. 151	Metals; Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property

FSC 3110

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

United States of America Standards Institute

AFBMA - Standards USA B46.1 - 1962 — Sections 3, 4, 10, 13 Surface Texture (Surface Roughness, Waviness and Lay)

(Copies of the above publications may be obtained from the United States of America Standards Institute, 10 East 40th Street, New York, N. Y. 10016, and the Anti-Friction Bearing Manufacturer's Association, Inc., 60 East 42nd Street, New York, N. Y. 10017.)

Uniform Classification Committee

Uniform Freight Classification Rules

(Application for copies of the above publication should be addressed to the Uniform Classification Committee, 202 Chicago Union Station, Chicago, Ill. 60606.)

3. REQUIREMENTS

3.1 This specification makes provisions for preproduction inspection (see 4.3 and 6.3).

3.2 Preproduction samples.- Prior to initiating production, samples shall be examined and tested as specified herein. Samples shall be forwarded for testing as specified in the contract or order (see 6.2).

3.3 Materials.-

3.3.1 Ball bearing rings and balls shall be made of AMS6444 or FS51100 steel in accordance with MIL-S-7420 or AMS6490 steels in accordance with QQ-S-624 and Fed. Std. No. 66. Material shall be CEVM steel or equal, shepherd grain size shall be 8 or finer. The manufacturer shall comply with the requirements of Fed. Std. No. 66 and Fed. Std. No. 151 to assure the production of bearings of the highest quality.

3.3.2 Case hardened or work hardened balls are not acceptable.

3.3.3 The ball bearing assemblies shall be manufactured from new component parts only, and in accordance with Section 3, Grade ABEC 7 of Anti-Friction Bearing Manufacturer's Association standard.

3.4 Design and construction.- The bearings shall be of such design and construction as to meet the requirements specified herein. Except as otherwise specified in the contract or order, the details of the working parts shall be at the option of the contractor.

3.4.1 Rings.-The inner and outer rings and balls shall be through hardened within the range of Rockwell C 60-64.

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3.4.2 Balls.- Chrome alloy steel balls shall fully satisfy the requirements of Section 10 of the Anti-Friction Bearing Manufacturer's Association standard for grade 10 or better and MIL-B-1083.

3.4.3 Separators.- Ball bearings shall, unless otherwise specified in the contract or order, have separators and shall be of material, workmanship, and finish as specified by the end item manufacturer. Separators shall not be adversely affected by lubricants and preservatives. The separator shall be of such a design as to admit the lubricant freely to the components of the bearing.

3.5 Dimensions.- Dimensions shall be as shown in table I.

3.6 Radial internal clearance.- The internal radial clearance shall be as specified by the end item manufacturer.

3.7 Surface finish.- Surface finishes shall be in accordance with the values specified in table II. Bearings shall be free of any defects which may be detrimental to satisfactory installation, performance or bearing life as defined by this specification.

3.8 Stability.- The bearing components shall be dimensionally stable for operation from -67° F to 350° F, unless otherwise specified by the end item manufacturer.

3.9 Vibration.- Bearings shall meet the vibration requirements specified in table III.

3.10 Interchangeability.- All bearing assemblies having the same part number shall be completely interchangeabile with each other in respect to installation and performance.

3.11 Identification of product.- The bearings shall be permanently and legibly marked for identification in accordance with MIL-STD-130. The identification data shall be as follows: Complete manufacturer's part number and name or trademark.

3.12 Workmanship.- The workmanship shall be in accordance with high-grade aircraft bearings manufacturing practice. The bearings shall be free from tool marks, chatter waves, rust, grinding scratches, pits, or any other defects that may adversely affect their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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TABLE I. Boundary dimensions

Bearing bore diameter		Bearing outside diameter		Bearing ring width		Inner ring radial runout	Outer ring radial runout	Width variation of individual ring		Inner ring side runout with bore	Outside cylindrical surface runout with side	Inner ring raceway runout with side	Outer ring raceway runout with side
								Inner	Outer				
Mm	Inches	Mm	Inches	Mm	Inches	Max	Max	Max	Max	Max	Max	Max	Max
10	0.3937	22	0.8661	6	0.2362	.0001	.00015	.0001	.0001	.0001	.00015	.0001	.0002
10	.3937	26	1.0236	8	.3150								
10	.3937	30	1.1811	9	.3543								
10	.3937	35	1.3780	11	.4331								
12	.4724	24	0.9449	6	.2362	.0001	.00015	.0001	.0001	.0001	.00015	.0001	.0002
12	.4724	28	1.1024	8	.3150								
12	.4724	32	1.2598	10	.3937								
12	.4724	37	1.4567	12	.4724								
15	.5906	28	1.1024	7	.2756	.0001	.00015	.0001	.0001	.0001	.00015	.0001	.0002
15	.5906	32	1.2598	9	.3543								
15	.5906	35	1.3780	11	.4331								
15	.5906	42	1.6535	13	.5118								
17	.6693	30	1.1881	7	.2756	.0001	.00015	.0001	.0001	.0001	.00015	.0001	.0002
17	.6693	35	1.3780	10	.3937								
17	.6693	40	1.5748	12	.4724								
17	.6693	47	1.8504	14	.5512								
20	.7874	37	1.4567	9	.3543	.0001	.00015	.0001	.0001	.00015	.00015	.00015	.0002
20	.7874	42	1.6535	12	.4724								
20	.7874	47	1.8504	14	.5512								
20	.7874	52	2.0472	15	.5906								
25	.9843	42	1.6535	9	.3543	.0001	.00015	.0001	.0001	.00015	.00015	.00015	.0002
25	.9843	47	1.8504	12	.4724								
25	.9843	52	2.0472	15	.5906								
25	.9843	62	2.4409	17	.6693								
30	1.1811	47	1.8504	9	.3543	.0001	.00015	.0001	.0001	.00015	.00015	.00015	.0002
30	1.1811	55	2.1654	13	.5118								
30	1.1811	62	2.4409	16	.6299								
30	1.1811	72	2.8346	19	.7480								
35	1.3780	55	2.1654	10	.3937	.00015	.0002	.0001	.0001	.00015	.00015	.00015	.0002
35	1.3780	62	2.4409	14	.5512								
35	1.3780	72	2.8346	17	.6693								
35	1.3780	80	3.1496	21	.8268								
40	1.5748	62	2.4409	12	.4724	.00015	.0002	.0001	.0001	.00015	.00015	.00015	.0002
40	1.5748	68	2.6772	15	.5906								
40	1.5748	80	3.1496	18	.7087								
40	1.5748	90	3.5433	23	.9055								

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TABLE II. Surface finish in microinches AA (10^{-6} in)

Bore size (mm)	Cross race	Guiding lands
10	10	32
12	10	32
15	10	32
17	10	32
20	10	32
25	15	32
30	15	32
35	15	32
40	15	32

TABLE III. Vibration requirements

Bore size (mm)	Low anderons	Medium anderons	High anderons
10	32	24	25
12	32	24	25
15	40	32	40
17	40	32	40
20	40	32	40
25	50	40	40
30	50	40	60
35	50	40	60
40	60	50	60

4.1.1 Preproduction test records.- The manufacturer shall maintain a record showing quantitative and qualitative results for all test required by this specification. This record shall be available to the purchaser and shall be signed by an authorized representative of the manufacturer or the testing laboratory, as applicable.

4.2 Classification of inspection.- The examination and testing of the bearings shall be classified as:

(a) Preproduction inspection (4.3)
(b) Quality conformance inspection (4.4)

4.3 Preproduction inspection.-

4.3.1 Tests.- The preproduction inspection shall consist of all the tests specified under 4.5. The tests shall be performed as specified in table IV.

4.3.2 Preproduction inspection samples.- The samples shall be plainly identified by securely attached durable tags or labels marked with the following information and shall be forwarded as specified in 6.2.

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Sample for preproduction inspection
BEARING, BALL, ANNULAR, ANTIFRICTION, ACCESSORY, HIGH SPEED
TURBINE, AIRCRAFT
MIL-B-83175
Name of manufacturer
Product code number
Date of manufacture
Contract or order number

4.4 Quality conformance inspection.- The quality conformance inspection shall consist of individual tests and sampling tests.

4.4.1 Lot.- A lot shall consist of bearings of one type, material, and sizes offered for delivery at one time.

4.4.2 Sampling.- The sampling bearings shall be selected in accordance with MIL-STD-105, inspection level, acceptable quality level 1.0. Inspection for compliance with 4.4.3 shall be 100 percent. In addition, the bearings may be subjected to any other tests specified herein considered necessary to determine conformance with this specification.

4.4.3 Individual tests.- All bearings shall be subjected to the following tests.

(a) Examination of product
(b) Radial internal clearance
(c) Noise level vibration

4.5 Inspection methods.-

4.5.1 Examination of product.- Each bearing shall be carefully examined to determine conformance with this specification as to dimensions, tolerances, lubrication and workmanship. In addition, all bearings shall be inspected for vibration and noise in accordance with AFBMA Standard,Section No. 13. Test values are specified in table III.

4.5.2 Inspection of preparation for delivery.- Preservation, packaging, packing and marking shall be inspected to determine conformance to section 5. A minimum of six bearings shall be selected at random for this examination.

4.5.3 Hardness.- Two bearings of each size shall be disassembled and tested for hardness of the inner race and outer race. Tests shall be made with a standard hardness tester using the Rockwell "C" scale. Tests shall be made on flat surfaces ground on rolling contact areas.

4.5.4 Balls.- Three balls from each bearing disassembled as specified in 4.5.3 shall be tested in accordance with USA B46.1.

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TABLE IV. Preproduction inspection samples

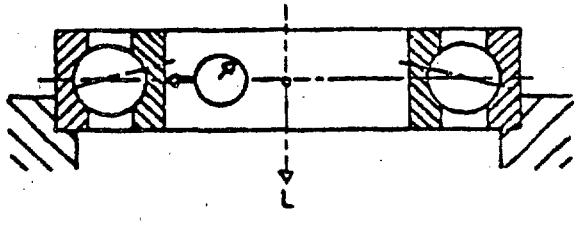
Examination of test	Sample number																				Test paragraph
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	
Examination of product	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	4.5.1
Radial internal clearance	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	4.5.5
Preservation, packaging, packing and marking	X	X	X	X	X	X															4.5.2
Radial runout (inner ring)	X	X	X	X	X	X	X	X	X	X											4.5.6
Radial runout (outer ring)											X	X	X	X	X	X	X	X	X	X	4.5.7
Width variation (inner ring)	X	X	X	X	X	X	X	X	X	X											4.5.8
Width variation (outer ring)											X	X	X	X	X	X	X	X	X	X	4.5.9
Side runout with bore	X	X	X	X	X	X	X	X	X	X											4.5.10
Raceway runout with side (outer ring)											X	X	X	X	X	X	X	X	X	X	4.5.11
Raceway runout with side (inner ring)	X	X	X	X	X	X	X	X	X	X											4.5.12
Outside surface runout with side											X	X	X	X	X	X	X	X	X	X	4.5.13
Hardness	X	X	X																		4.5.3
Surface roughness	X	X	X																		4.5.14
Stabilization	X	X	X																		4.5.15
Separators				X	X	X	X	X	X	X	X	X	X								4.5.16

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4.5.5 Radial internal clearance.- The bearing shall be mounted with its axis in a horizontal position and one of its rings supported to prevent movement. The unsupported ring shall be permitted to float both axially and radially while the specified load of 5-1/2 pounds moves this member upward and downward. Radial clearance shall be the total dimetrical movement of the unsupported ring when the specified load is reversed, and shall be the average of readings taken at different relative angular positions of the inner rings, outer rings, and balls, with sufficient travel between readings to provide a full rotation of the set of balls.

4.5.6 Radial runout (inner ring) (figure 1).- Method applicable to groove type ball bearings (including single row angular contact).

4.5.6.1 Support the outer ring and apply a small dynamically stable co-axial load to the reference or thrust side of the inner ring to ensure contact between balls and raceways. Apply an indicator to the center line of the inner ring bore and take indicator readings while rotating the inner ring one revolution.



LOAD ON INNER RING

FIGURE 1. Radial runout (inner ring)

4.5.7 Radial runout (outer ring) (Figure 2).- Method applicable to groove type ball bearings (including single row angular contact.)

4.5.7.1 Support the inner ring and apply a small dynamically stable co-axial load to the reference side or thrust side of the outer ring to ensure contact between balls and raceways. Apply an indicator to the middle of the outer cylindrical surface and take indicator readings while rotating the outer ring one revolution.

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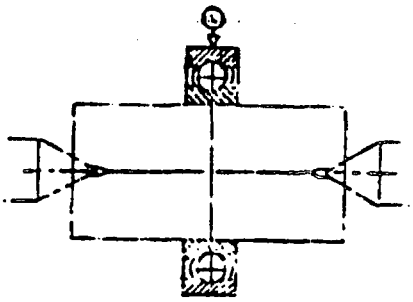


FIGURE 2. Radial runout (outer ring)

4.5.8 Width variation (inner ring)(figure 3).- Support one side of the inner ring on three buttons, leaving the outer ring free. Apply an indicator against the opposite side of the inner ring directly opposite one button. Take indicator readings while rotating the inner ring one revolution.

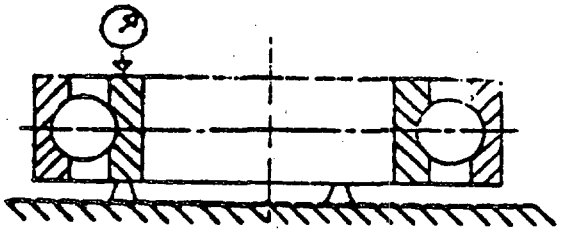


FIGURE 3. Width variation (inner ring)

4.5.9 Width variation (outer ring) (figure 4).- Support one side of the outer ring on three buttons, leaving the inner ring free. Apply an indicator against the opposite side of the ring directly opposite one button. Take indicator readings while rotating the outer ring one revolution.

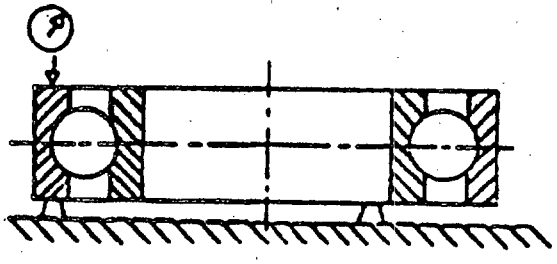


FIGURE 4. Width variation (outer ring)

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4.5.10 Side runout with bore (figure 5).- Mount the bearing on an arbor having a taper of 0.0001 to 0.0002 inch on diameter per inch of length and apply the arbor between two accurate centers so that it can be rotated. Apply an indicator against the reference side of the inner ring on a diameter as near equal to the mean raceway diameter as possible. Take indicator readings while rotating the inner ring one revolution. Correction must be made for inaccuracy of the arbor.

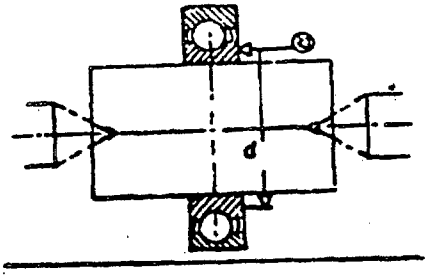


FIGURE 5. Side runout with bore

4.5.11 Raceway runout with side (outer ring) (figure 6).- Support the inner ring and apply a small dynamically stable co-axial load to the outer ring to ensure contact between the balls and the raceways. Apply an indicator on the reference or thrust side of the outer ring and take indicator readings while rotating the outer ring one revolution.

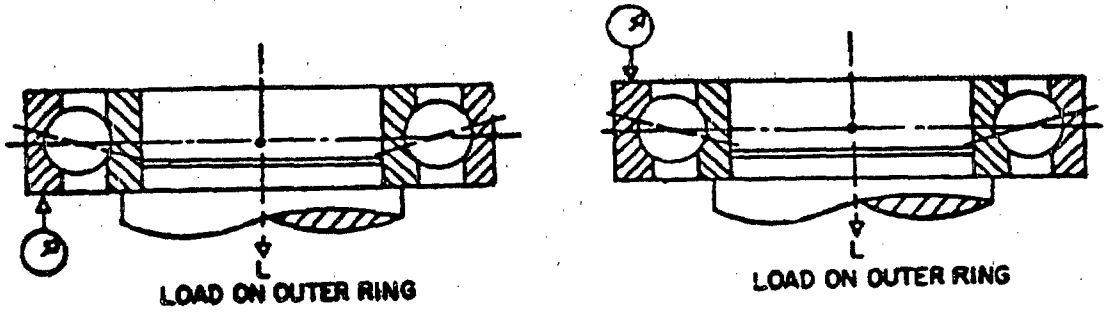


FIGURE 6. Raceway runout with side (outer ring)

4.5.12 Raceway runout with side (inner ring)(figure 7).- Support the outer ring and apply a small dynamically stable co-axial load to the inner ring to assure contact between the balls and the raceways. Apply an indicator on the reference side of the inner ring and take indicator readings while rotating the inner ring one revolution.

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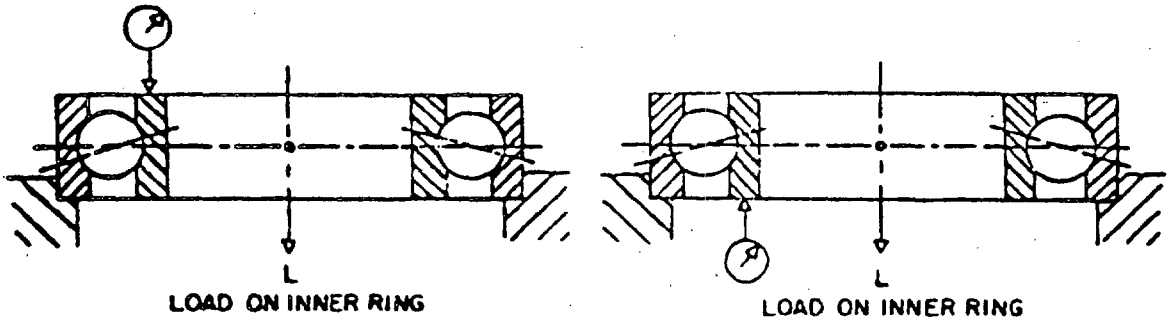


FIGURE 7. Raceway runout with side (inner ring)

4.5.13 Outside surface runout with side (figure 8).- Support the reference side of the outer ring, leaving the inner ring free, and locate the outside cylindrical surface against a stop a distance of 3.5r from the lower side surface. Apply an indicator directly above the sop at a distance of 3.5r from the upper side surface illustrated. Take indicator readings while rotating the inner ring one revolution.

r = maximum housing fillet radius which the bearing chamber must clear

(see AFBMA Standard Section No. 2, table I).

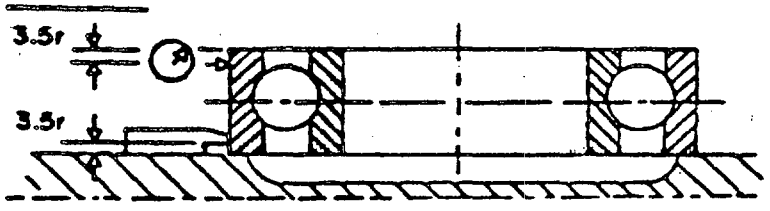


FIGURE 8. Outside surface runout with side

4.5.14 Surface roughness.- Surface roughness shall be measured in accordance with USA B46.1. Bearing parts on which the hardness checks of 4.5.3 were made may be used for this measurement.

4.5.15 Stabilization.- The bearing manufacturer shall show evidence that the heat treatment process used for rings and balls will meet the following requirements. Bearing rings and balls, measured at room temperature, shall not show dimensional change of more than 0.0001 inch per inch of diameter after being held for 100 hours at the operating temperature specified by the end item manufacturer.

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4.5.16 Separators.- There shall be no cracks on machined metallic cages visible by fluorescent penetrant inspection. Inspections shall be in accordance with MIL-I-6866. On nonmetallic cages there shall be no cracks or delaminations visible at 5X magnification. There shall be no separation of metal and nonmetal in metal bonded nonmetallic cages when inspected by 5X magnification (see figure 9)

4.5.16.1 There shall be no circumferential mismatch of separator halves on two-piece separators in excess of 0.004 inch as shown on figure 10.

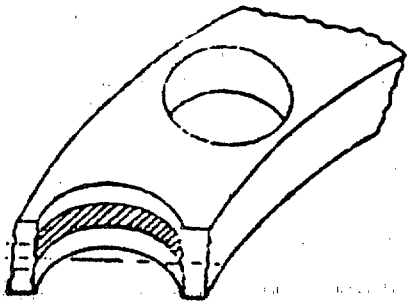


FIGURE 9. View of separators

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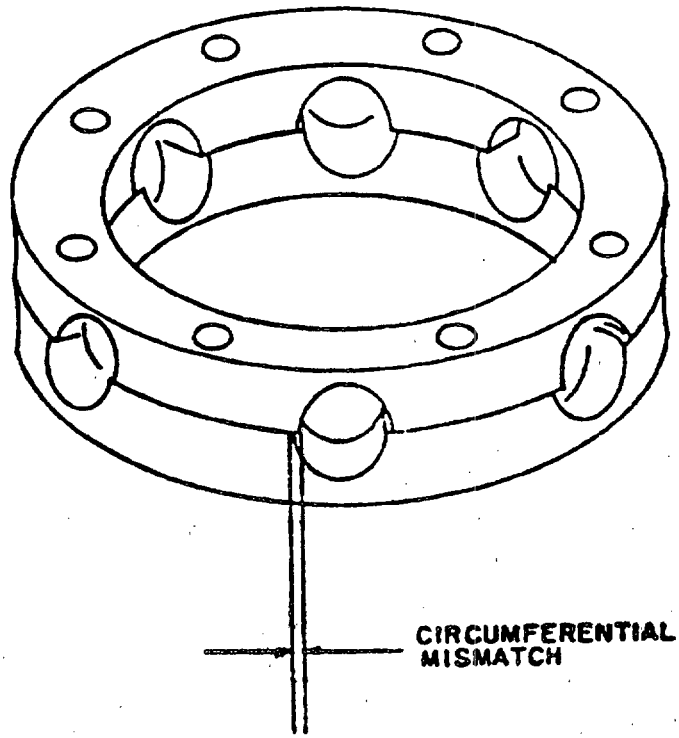


FIGURE 10. View of circumferential mismatch of separators

4.5.16.2 There shall be no radial mismatch of separator halves on two-piece separators in excess of 0.004 inch, as measured on guiding surface as shown on figure 11.

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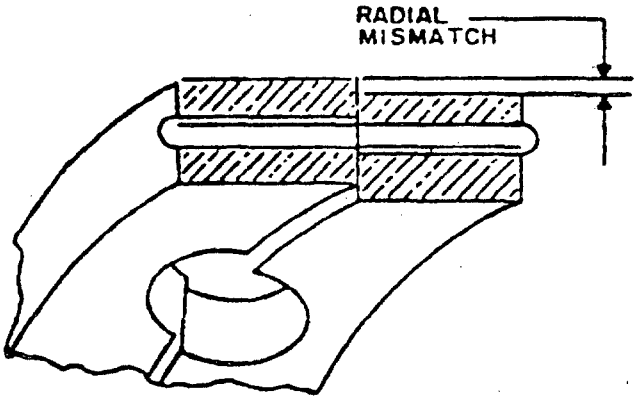


FIGURE 11. View of radial mismatch of separators

4.5.16.3 There shall be no cracks or separators visible by magnaflux or zyglo, or any crack or lamination by 5 X magnification. Inspection shall be in accordance with MIL-I-6866 and MIL-I-6868. There shall be no separation of metal and phenolic in metal bonded phenolic separators as seen by 5 X magnification.

Separators to be round within:	Over		Inclusive
.002 inch on bearing with bore	0	to	19 mm
.004 inch on bearing with bore	19	to	39 mm
.006 inch on bearing with bore	39	to	80 mm

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Bearings shall be cleaned, preserved and packaged in accordance with level A of MIL-B-197.

5.2 Packing.- Bearings, preserved and packaged as specified in 5.1 shall be packed in accordance with level A, B,or C of MIL-B-197, as specified (see 6.2).

5.3 Marking of shipments.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall include: BEARING, BALL, ANNULAR ANTIFRICTION, ACCESSORY, HIGH SPEED TURBINE, AIRCRAFT

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6. NOTES

6.1 Intended use.- The bearings covered by this specification are intended primarily for use in aircraft accessories. The loads are relatively low and speeds are very high. The bearings are usually lubricated by engine lubricants and operate near engine temperatures.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Complete part number.
(c) Levels of packing (see 5.2).
(d) Where preproduction samples are to be forwarded for testing.

6.3 Preproduction.- Unless otherwise specified, the activity responsible for the preproduction testing is the Aeronautical Systems Division, Wright-Patterson Air Force Base, Ohio 45433, and information pertaining to preproduction inspection may be obtained from that activity.

6.4 In order to procure replacement bearings qualified to the original end item equipment specification, bearings should be purchased from the end item manufacturer or the bearing manufacturer.

Custodians:
Army - AV
Navy - SH
Air Force - 11

Reviewer activities:
Army - AV
Navy - SH
Air Force - 82, 71

User activities:
Navy - AS

Preparing activity:
Air Force - 11

Project No. 3110-0315

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget
Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION MIL-B-83175 BEARING, BALL, ANNULAR, ANTIFRICTION, ACCESSORY, HIGH SPEED TURBINE AIRCRAFT

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A
( ) Direct Government Contract ( ) Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE?
   ( ) YES ( ) NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity) DATE

DD Form 1426

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